

Hanvit Cho

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As a Master's student specializing in robotics at Stanford University, I have developed a solid foundation in **designing, controlling, and integrating** various robotic systems—including manipulators, drones, and swarm robots. My expertise is further enhanced by hands-on experience in **imitation learning** and **PID controller implementation** for a drone project during my undergraduate studies. With a strong background in **mechanical design** and validating **electrical components and system functionalities**, I am well-prepared to contribute to the **optimization and reliability** of complex sub-systems within autonomous platforms.

EDUCATION

Stanford University | Stanford, California

September 2023 – June 2025

- MS in Mechanical Engineering, **Concentration in Robotics**.

University at Buffalo - State University of New York (SUNY) | Buffalo, New York

August 2021 – May 2023

- BS in Mechanical Engineering. GPA: 4.0/4.0

WORK EXPERIENCE

Robotics Internship

Centrillion Technology

Full-time intern

June 2024 – September 2024

- **Developed and optimized robotic solutions** using **C++** and **Python** by integrating Trossen Robotics' Mobile ALOHA through advanced **imitation learning** techniques, improving manufacturing automation in real-world situations.
- Implemented hardware-in-the-loop testing, refined **control algorithms**, and validated diagnostic protocols to ensure reliable robotic operations under highly variable conditions.

Sun-Signs Engineering Internship

University at Buffalo

Part-time intern

February 2022 – July 2022

- Designed and assembled a solar-powered navigation device using **SOLIDWORKS** and utilized **3D printing** to create a prototype.
- Conducted failure analysis using sensor data in **MATLAB**, refining mechanical integrity and overall system resilience.
- Evaluated functionality in diverse environmental conditions to enhance durability, **iterating design decisions** based on data-driven optimization for autonomous operation.

RESEARCH EXPERIENCE

Stanford Vision and Learning Lab

Stanford University

Graduate student research assistant

June 2024 – June 2025

- Developed a **brain-robot interface** using **EEG signals** and machine learning pipelines for **real-time control** of robotic arms, improving system intelligence in semi-structured pick-place-pack tasks.
- Applied machine learning to enhance object detection accuracy, reducing classification errors under variable lighting.
- Developed test setups to enhance diagnostic capabilities, ensuring efficient robotic operations with **real-world applications**.

- o **NOIR 2.0: Neural Signal Operated Intelligent Robots for Everyday Activities**

CoRoboLearn 2024

Kim, T., Wang, Y., Cho, H., Hodges, A.

Collaborative Haptics and Robotics in Medicine Lab

Stanford University

Graduate student research assistant

March 2024 – July 2024

- Redesigned an inflatable **vine-like robot** with improved sealing mechanisms and servo-based pressure control, minimizing leakage and enhancing maneuverability in constrained spaces by utilizing **GD&T**.
- Employed **FEA** to refine structural design, improving reliability under various dynamic loads.

Salisbury Robotics Lab

Stanford University

Graduate student research assistant

March 2024 – July 2024

- Contributed to developing robotic Emergency Medical Technician (rEMT) using Kinova Gen 3 arm, Bota force sensor, and Haply Inverse-3 haptic device for precise medical palpation and diagnostics.
- Programmed **ROS-based motion control algorithms** and **haptic perception** loops to enhance force feedback accuracy, ensuring precise diagnostic and therapeutic interventions.
- Led the **testing phase**, which involved rigorous trials to evaluate the performance aspects, ensuring the system's reliability.

Adaptive Design Algorithms, Models & Systems Lab

University at Buffalo

Undergraduate student research assistant

February 2022 – June 2023

- Developed **real-time control** algorithms in **C++** and **Python** for e-puck2 ground robots and Crazyflie drones, emphasizing robust sensor fusion and rapid trajectory adjustments.
- Utilized ROS 2 frameworks with Vicon motion capture for **closed-loop control experiments**, refining swarm coordination for precise object manipulation and automated collaboration.
- Streamlined data acquisition and feedback loops, improving response times and control accuracy by integrating **real-time telemetry** and sensor data across multiple robotic agents.

- o **Fast Decision Support for Air Traffic Management at Urban Air Mobility Vertiports Using Graph Learning**

IROS 2023

KrishnaKumar, P., Witter, J., Paul, S., Cho, H., Dantu, K., Chowdhury, S.

- o **Framework for Analyzing Human Cognition in Operationally-Relevant Human Swarm Interaction**

ASME 2023

Distefano, J., Cho, H., KrishnaKumar, P., Esfahani, E., Chowdhury, S.

COMPUTATIONAL SKILLS

Languages: C (Expert), C++ (Expert), Python (Expert), Java (Intermediate), JavaScript (Intermediate)

Applications: ROS2 (Expert), MATLAB (Expert), SOLIDWORKS (Expert), Adobe Inventor (Expert)

Engineering Skills: Manual Machining, GD&T, FEA, Motion Capture (Vicon Tracker), EEG Brain Signal (NetStation)